

123456

Sheet 1

File: memory-v2.0-sheet1.kicad_sch

Sheet 6

File: memory-v2.0-sheet6.kicad_sch

Sheet 2

File: memory-v2.0-sheet2.kicad_sch

Sheet 7

File: memory-v2.0-sheet7.kicad_sch

Sheet 3

File: memory-v2.0-sheet3.kicad_sch

Sheet 4

File: memory-v2.0-sheet4.kicad_sch

Sheet 5

File: memory-v2.0-sheet5.kicad_sch

Circuit source: mem_v2_netlist.py; generated by gen_mem_v2.py; see README.md
Sheets 1-6 = memory card v1.3 as built (sheet 1: six bus labels made global); sheet 7 = CompactFlash v2.0

Sheet: /
File: memory-v2.0.kicad_sch

Title: YACC1 memory card v2.0

Size: A4Date: 2026-09-24

Rev: v2.0

KiCad E.D.A. 10.0.6

Id: 1/8

654321

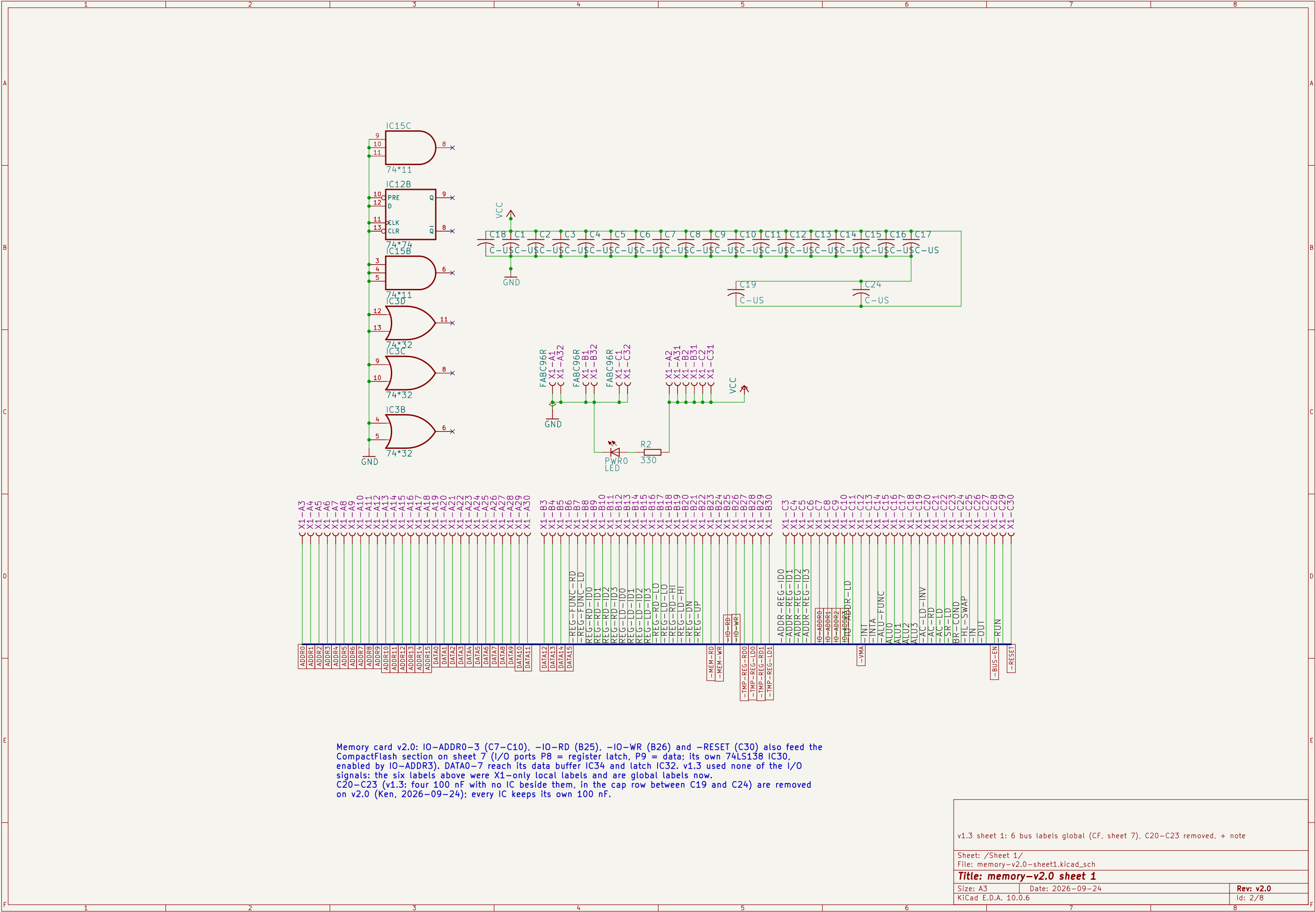
A

B

C

D

CompactF



Memory card v2.0: IO-ADDR0-3 (C7-C10), -IO-RD (B25), -IO-WR (B26) and -RESET (C30) also feed the CompactFlash section on sheet 7 (I/O ports P8 = register latch, P9 = data; its own 74LS138 IC30, enabled by IO-ADDR3). DATA0-7 reach its data buffer IC34 and latch IC32. v1.3 used none of the I/O signals: the six labels above were X1-only local labels and are global labels now. C20-C23 (v1.3: four 100 nF with no IC beside them, in the cap row between C19 and C24) are removed on v2.0 (Ken, 2026-09-24); every IC keeps its own 100 nF.

v1.3 sheet 1: 6 bus labels global (CF, sheet 7), C20-C23 removed, + note

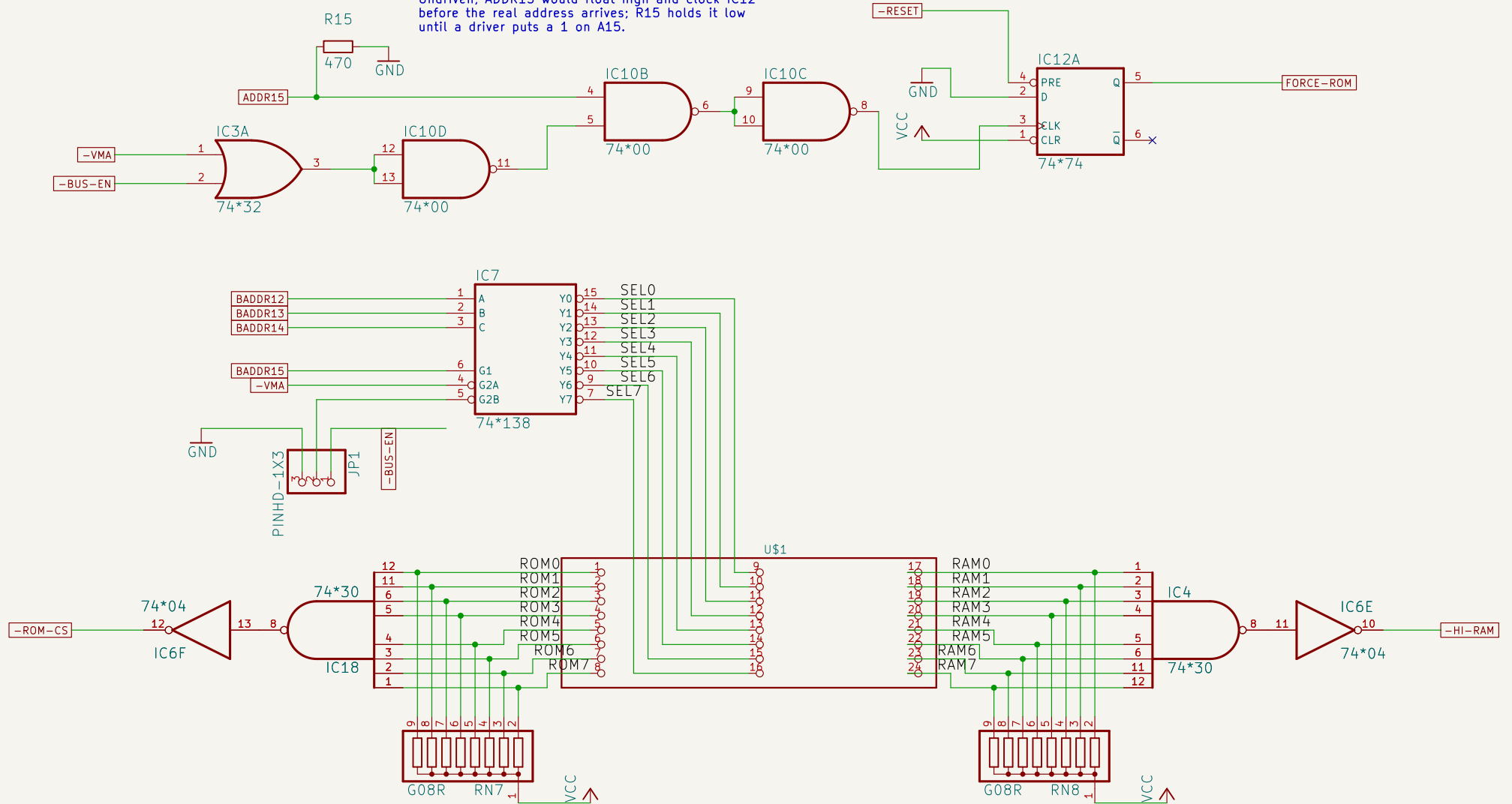
Sheet: /Sheet 1/
File: memory-v2.0-sheet1.kicad_sch

Title: memory-v2.0 sheet 1

Size: A3 Date: 2026-09-24
KiCad E.D.A. 10.0.6

Rev: v2.0
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R15 470: ADDR15 pull-down, FORCE-ROM race fix (review M1, 2026-09-30); docs/cards/memory.md 4.1 Undriven, ADDR15 would float high and clock IC12 before the real address arrives; R15 holds it low until a driver puts a 1 on A15.



v1.3 sheet 2 + R15, the ADDR15 pull-down: the FORCE-ROM race fix (rule 8, 2026-09-30)

Sheet: /Sheet 2/
File: memory-v2.0-sheet2.kicad_sch

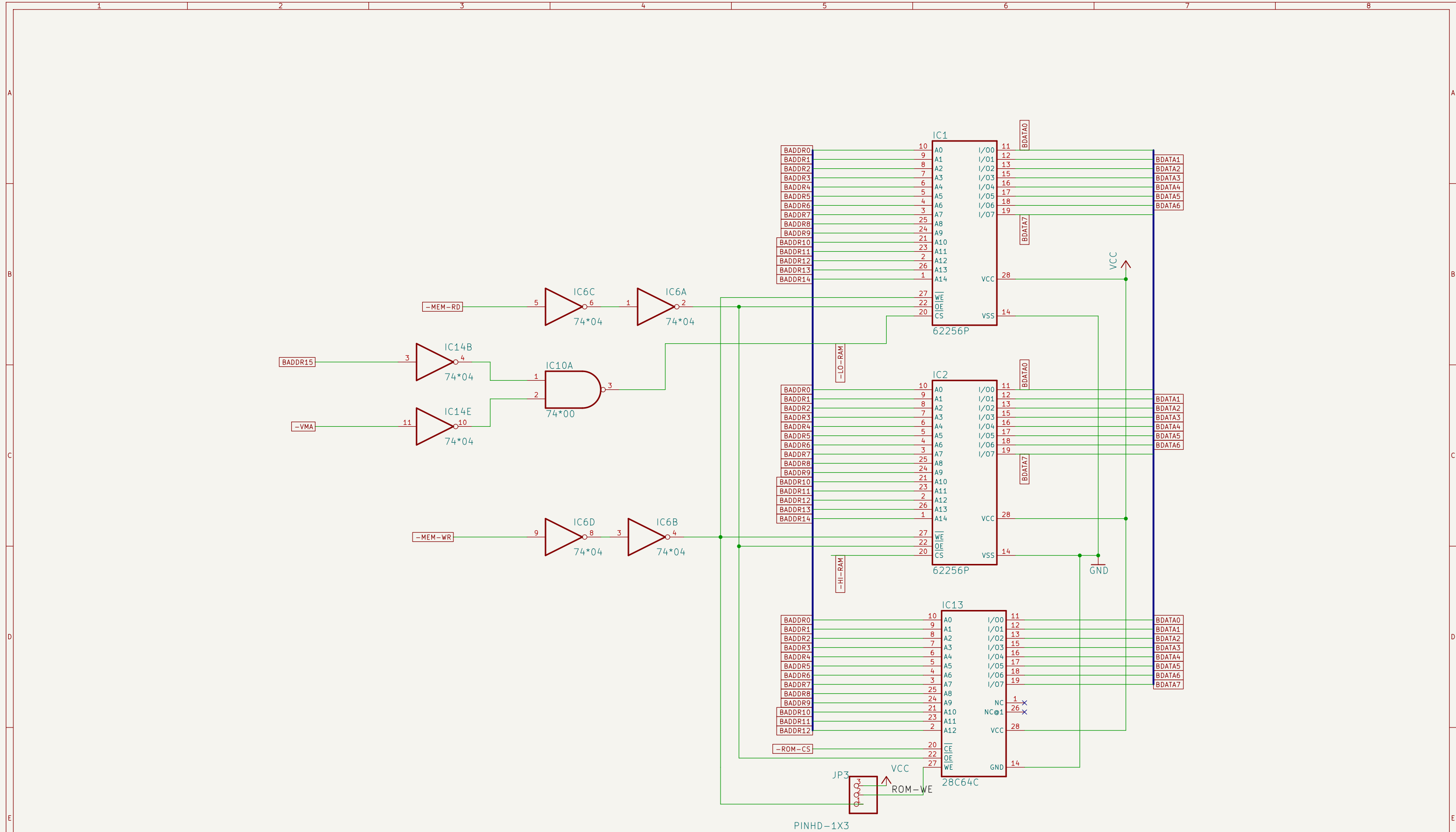
Title: memory-v2.0 sheet 2

Size: A4 Date: 2026-09-24

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Rev: v2.0

Id: 3/8



JP3 (v2.0, Ken 2026-09-29): ROM write protect (design review M2).
1-2 = WRITE: IC13 -WE = -MEM-WR, as on v1.3. 2-3 = PROTECT: -WE held at VCC.
Fitted on 2-3: burn the 28C64 in a programmer; 1-2 only to write it in place.

v1.3 sheet 3 + JP3, the ROM write-protect jumper on IC13 -WE (rule 7, 2026-09-29)

Sheet: /Sheet 3/
File: memory-v2.0-sheet3.kicad_sch

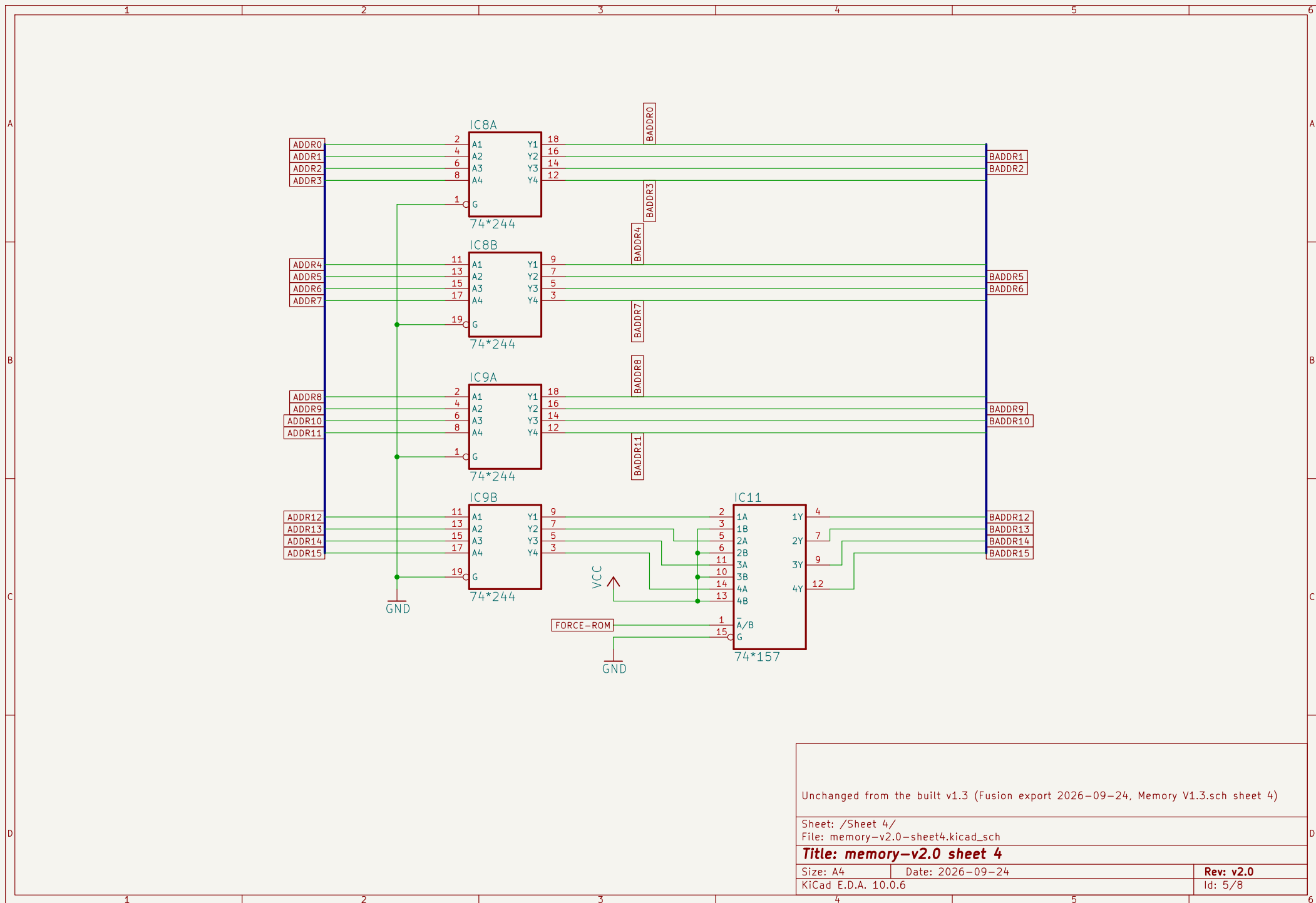
Title: memory-v2.0 sheet 3

Size: A3 Date: 2026-09-24

Rev: v2.0

KiCad E.D.A. 10.0.6

Id: 4/8



Unchanged from the built v1.3 (Fusion export 2026-09-24, Memory V1.3.sch sheet 4)

Sheet: /Sheet 4/
File: memory-v2.0-sheet4.kicad_sch

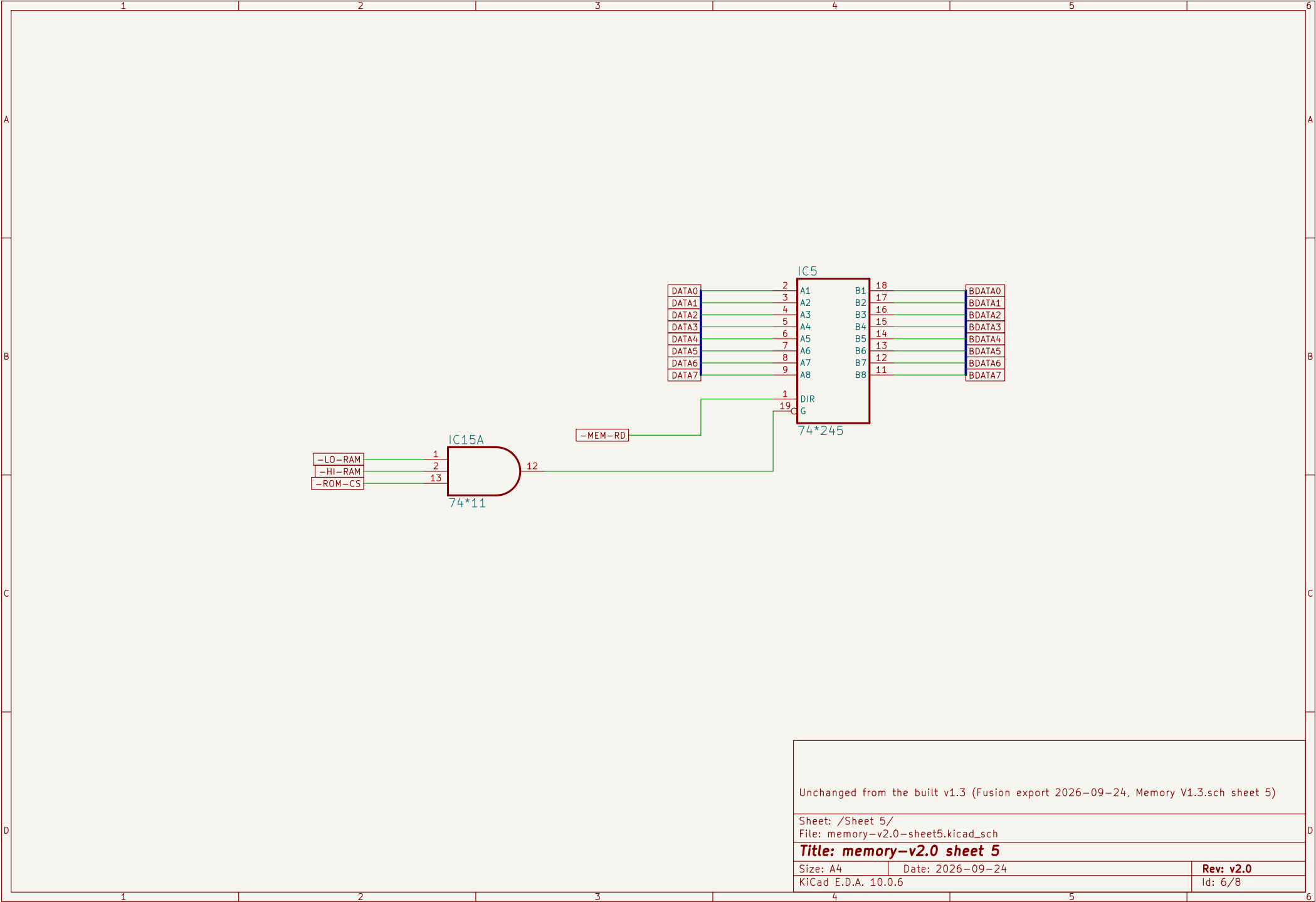
Title: memory-v2.0 sheet 4

Size: A4
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: v2.0

Id: 5/8



Unchanged from the built v1.3 (Fusion export 2026-09-24, Memory V1.3.sch sheet 5)

Sheet: /Sheet 5/
File: memory-v2.0-sheet5.kicad_sch

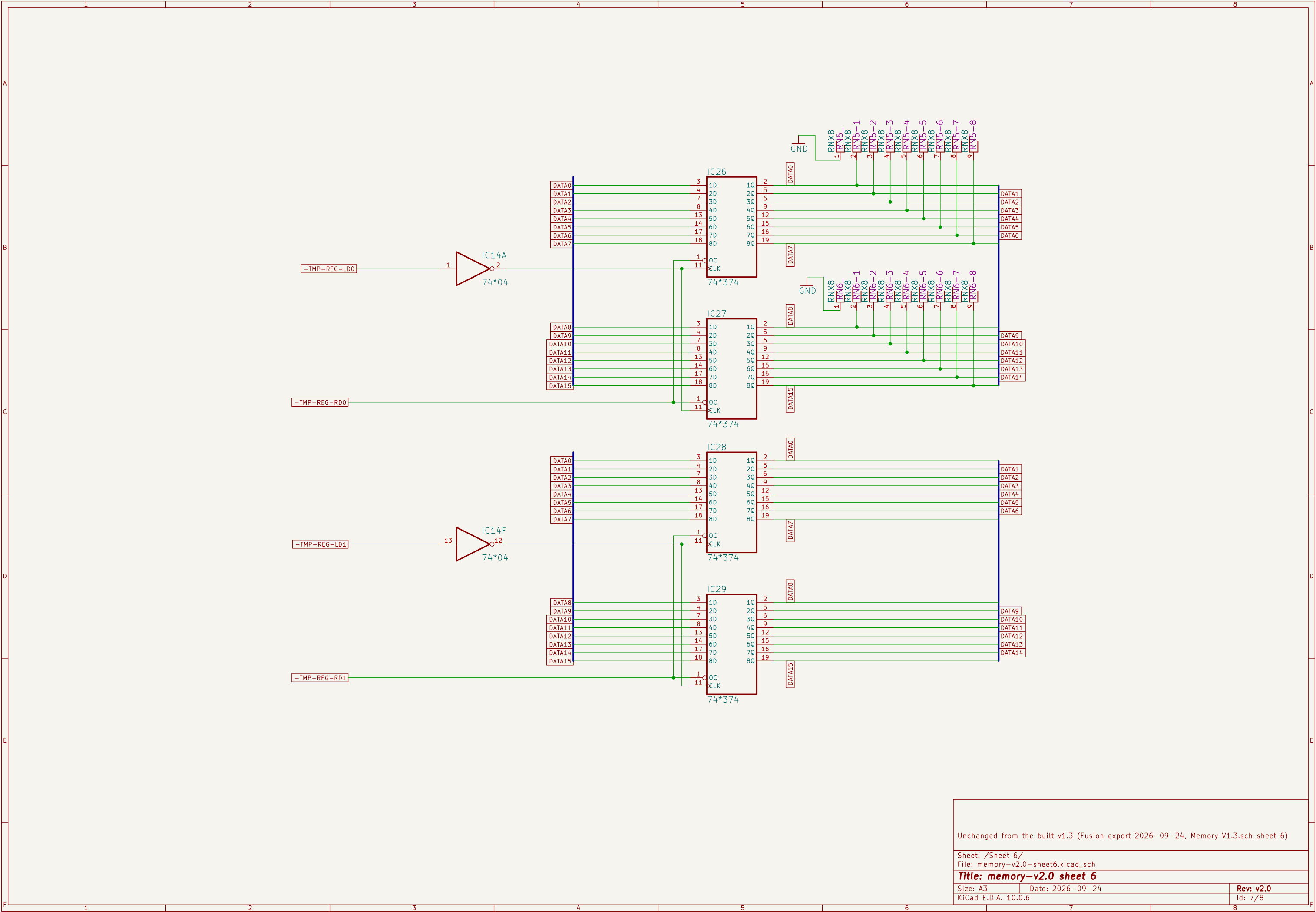
Title: memory-v2.0 sheet 5

Size: A4 Date: 2026-09-24

KiCad E.D.A. 10.0.6

Rev: v2.0

Id: 6/8



Unchanged from the built v1.3 (Fusion export 2026-09-24, Memory V1.3.sch sheet 6)

Sheet: /Sheet 6/
File: memory-v2.0-sheet6.kicad_sch

Title: memory-v2.0 sheet 6

Size: A3
Date: 2026-09-24

Rev: v2.0
Id: 7/8

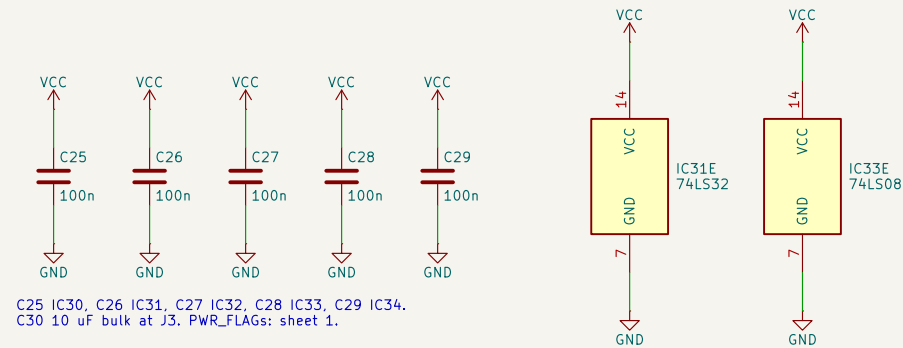
FROM THE BUS (memory card X1, sheet 1)

The CF section has no bus connector of its own; it shares the memory card's X1. Boxed global labels = nets shared with sheets 1–6:

IO-ADDR0-3 X1 C7-C10 port number -> IC30 (74LS138)
-IO-RD X1 B25 -> IC31 gate 2 (CF -IOR)
-IO-WR X1 B26 -> IC31 gates 1 and 3 (latch clock, CF -IOW)
-RESET X1 C30 -> IC32 CLR, IC33 gate 2 (also IC12 PRE)
DATA0-7 X1 A19-A26 -> IC34 A side; DATA0-3 -> IC32 D1-D4
VCC, GND the card's supply (six pins each on X1)

On v1.3, IO-ADDR0-3, -IO-RD and -IO-WR reached X1 only. The CF section drives the data bus only while the CPU reads port P9 (IC34 enabled by -CFOE, DIR = -IOR).

POWER + DECOUPLING (one 100 nF per new IC)

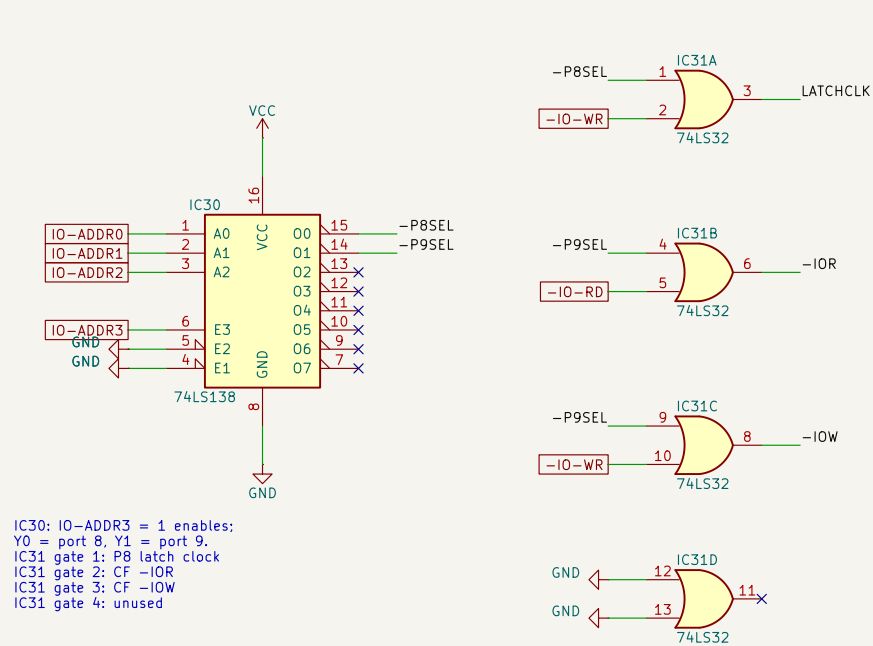


C25 IC30, C26 IC31, C27 IC32, C28 IC33, C29 IC34.
C30 10 uF bulk at J3. PWR_FLAGS: sheet 1.

DESIGN NOTES (MEMORY CARD v2.0, CF SECTION)

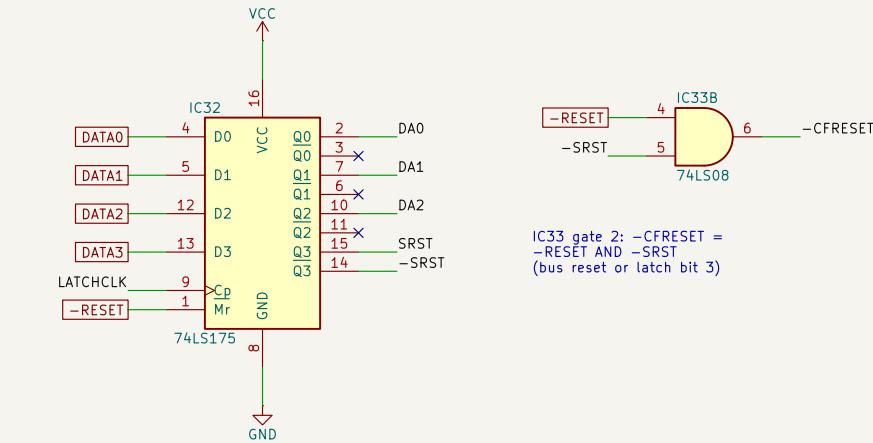
- Two I/O ports, P8 (write): 74LS175 latch IC32 – bits 0–2 = the ATA task-file register (CF DA0–2), bit 3 = CF reset (1 = held in reset). P9 (read/write): the selected ATA register, 8-bit True IDE, through the 74LS245 IC34.
 - Decode: 74LS138 IC30 on IO-ADDR0–2, enabled by IO-ADDR3 (Y0 = port 8, Y1 = port 9), as on the CF card v1.0. The ROM in the machine and both emulators use P8/P9. The I/O card must keep its IO-ADDR-HL strap at P0–P7.
 - IC31 (74LS32) ORs each port select with -IO-WR / -IO-RD: -IOR, -IOW and the latch clock are low only during the bus strobe for this port. -CS0 (J2 pin 37) LOW and -CS1 (38) HIGH: the strobes alone define every cycle (docs/cards/cf.md section 4: gating -CS0 would break the chip-select hold time after -IOR).
 - J2 = 40-pin IDE header (pin 20 kept) for a CF-to-IDE adapter: plugged straight on (TAODAN CF-IDE40, standing off the component side) or on a short ribbon (SinLoon). Adapter power: J3 (1 = +5 V, 2 and 3 = GND, 4 not used) or, with JP2 fitted, IDE pin 20 for adapters powered there.
 - Pull-ups: CF D0–7 (RN9: an empty adapter reads \$FF = 'no card' to the ROM driver), IORDY, -PDIAG, -DASP; -DMACK held inactive; CSEL grounded (master). SRST (IC32 Q4) is a one-pin net by design (a probe point).
 - From hardware/cards/cf/kicad/v1.0/cf_netlist.py: U1->IC30 U2->IC31 U3->IC32 U4->IC33 U5->IC34 J1->J2 J2->J3 JP1->JP2 RN1->RN9 R1-R5->R10-R14 LED2->LED1 C1-C6->C25-C30. Dropped: X1 (the card's bus connector), the PWR LED (LED1, R7: the card has PWR0 + R2) and the DASP LED (LED3, R6).
- Theory: docs/cards/cf.md, docs/cards/memory.md. Circuit source: mem_v2_netlist.py (edit, then re-run build.sh).

PORT DECODE + STROBE GATING



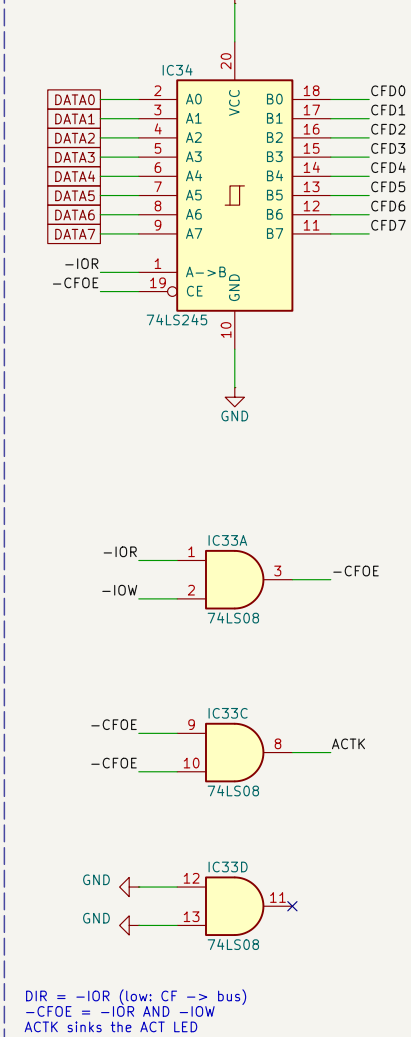
IC30: IO-ADDR3 = 1 enables:
Y0 = port 8, Y1 = port 9.
IC31 gate 1: P8 latch clock
IC31 gate 2: CF -IOR
IC31 gate 3: CF -IOW
IC31 gate 4: unused

P8 LATCH: DA0-2 + CF RESET



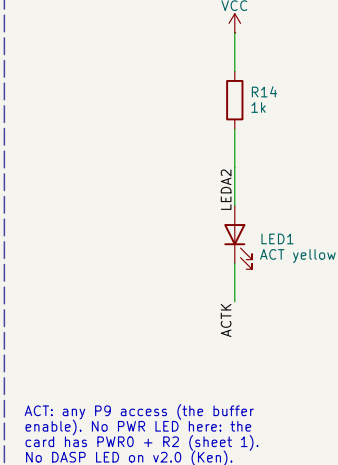
IC33 gate 2: -CFRESET =
-RESET AND -SRST
(bus reset or latch bit 3)

P9 DATA BUFFER



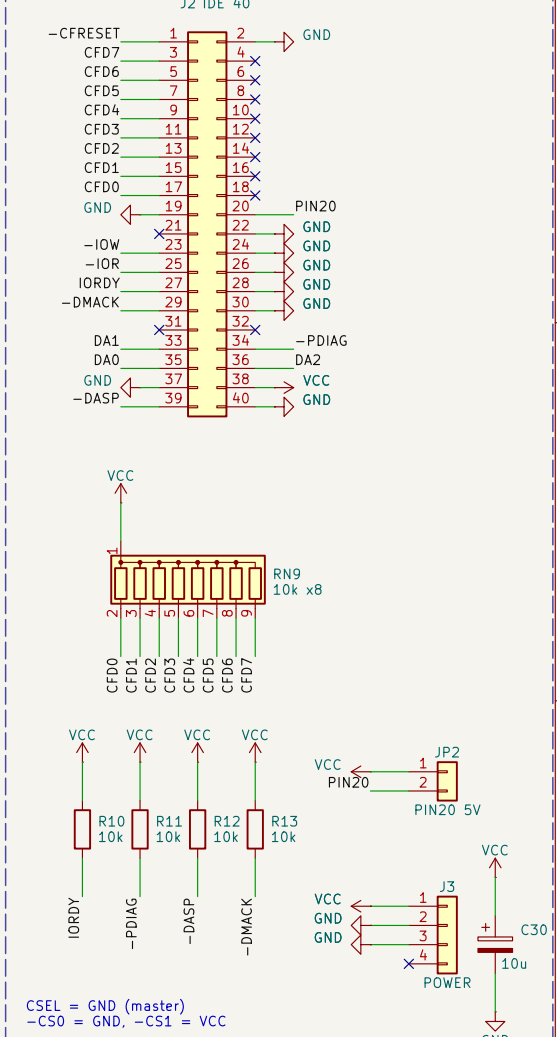
DIR = -IOR (low: CF -> bus)
-CFOE = -IOR AND -IOW
ACTK sinks the ACT LED

ACT LED



ACT: any P9 access (the buffer enable). No PWR LED here: the card has PWR0 + R2 (sheet 1). No DASP LED on v2.0 (Ken).

IDE HEADER J2 (CF-TO-IDE ADAPTER)



CSEL = GND (master)
-CS0 = GND, -CS1 = VCC